Modeling Damage Accumulation in Plasma-Facing Materials using the Norgett-Robinson-Torrens Model in ITER Conditions

Background:

At the forefront of next-generation energy, fusion reactors such as ITER seek to harness the same reactions that power the Sun to deliver sustainable and carbon-free electricity. Fusion reactors, like ITER, aim to use a Deuterium-Tritium plasma to undergo the fusion process to produce 14.1 MeV fast neutrons that are absorbed by the blanket materials. As these neutrons are slowed, their kinetic energy is transformed into heat, which is used to generate steam and drive turbines for electricity production. Despite their role in energy conversion, these fast neutrons also induce significant radiation damage on the Beryllium first-wall and the Tungsten divertor of the reactor. My project aims at modeling this damage accumulation within these materials under fusion conditions under ITER-relevant conditions.

Physics & Theory:

When fast neutrons interact with materials, they can transfer their kinetic energy to the atoms within the crystal structure and displace these atoms from their lattice site. These kick-out atoms are called primary knock-on atoms (PKA). These PKA then continue to move throughout the lattice and can collide with other atoms, creating a collision cascade. These cascades produce many defects within the crystal structure such as vacancies, voids, and interstitials. As a result of these deformities, the material undergoes degradation of its mechanical and thermal properties via embrittlement, swelling, hardening, thermal conductivity degradation, and surface erosion.

The Norgett-Robinson-Torrens Model is an internationally adopted standard model used to estimate the number of atomic displacements produced by irradiation, commonly expressed in terms of displacements per atom (DPA). It provides a simplified relationship between the damage energy transferred to the lattice and the resulting number of stable defects, based on an assumed displacement threshold energy and an empirical efficiency factor. While the model does not explicitly resolve the full complexity of collision cascades or defect recombination processes, it serves as a practical first-order metric for comparing radiation damage across materials and irradiation conditions. In fusion applications, the NRT model is widely used to quantify and benchmark neutron-induced damage in structural materials such as tungsten and beryllium,

providing a basis for advanced modelling approaches that account for cascade physics and temperature-dependent defect evolution.

Methods:

The irradiation environment is represented as a pulsed neutron flux, where the instantaneous flux $\phi(t)$ is defined as

$$\phi(t) = \begin{cases} \phi_0, & t \in \text{pulse} \\ 0, & t \in \text{dwell} \end{cases}$$

with a pulse duration of 450 s and a dwell period of 1380 s. The total neutron fluence incident on each material is computed at each time step using

$$N_n = \phi A \Delta t$$

where A is the exposed surface area and Δt is the temporal resolution of the simulation.

The number of atoms in each material volume is determined from

$$N = \frac{\rho V}{m_{atom}}$$

where ρ is the material density, V is the sample volume, and $m_{atom} = A/N_A$ is the atomic mass per atom derived from molar mass A and Avogadro's number N_A.

Energy transfer from incident neutrons to lattice atoms is treated as elastic two-body scattering, with the maximum transferable recoil (damage) energy given by

$$T_d = \frac{4 m_n m_A}{(m_n + m_A)^2} E_n$$

where m_n and m_A are the neutron and target atom masses, and E_n is the incident neutron energy (14.1 MeV for D–T fusion neutrons).

The resulting number of atomic displacements per neutron is calculated using the NRT model:

$$N_d = 0.8 \frac{T_d}{2E_d}$$

where E_d is the material-dependent displacement threshold energy (90 eV for tungsten and 25 eV for beryllium). An additional empirical correction factor $\kappa = 1/3$ is applied to account for defect recombination and cascade inefficiencies not captured in the original NRT formulation. At each time step, the total number of displacements is computed as the product of incident neutrons and the per-neutron damage production rate. The cumulative damage is then summed over time and normalized by the total number of atoms in each material to obtain the displacement per atom (dpa):

$$\mathrm{DPA}(t) = \frac{\sum_i N_{disp,i}}{N_{atoms}}$$

This framework enables direct comparison of radiation damage accumulation between tungsten and beryllium under identical ITER-like pulsed neutron loading conditions, while capturing the time-resolved nature of the irradiation environment.

Results:

The model predicts time-dependent accumulation of radiation damage (dpa) in both tungsten (W) and beryllium (Be) under ITER-like pulsed neutron irradiation, with distinct differences in magnitude and growth behavior between the two materials.

In both cases, the displacement per atom increases in a staircase-like manner, with damage accumulating during neutron pulse phases and remaining constant during dwell periods when the flux is zero. This reflects the direct dependence of defect production on the instantaneous neutron flux, resulting in stepwise increases in dpa aligned with each irradiation cycle.

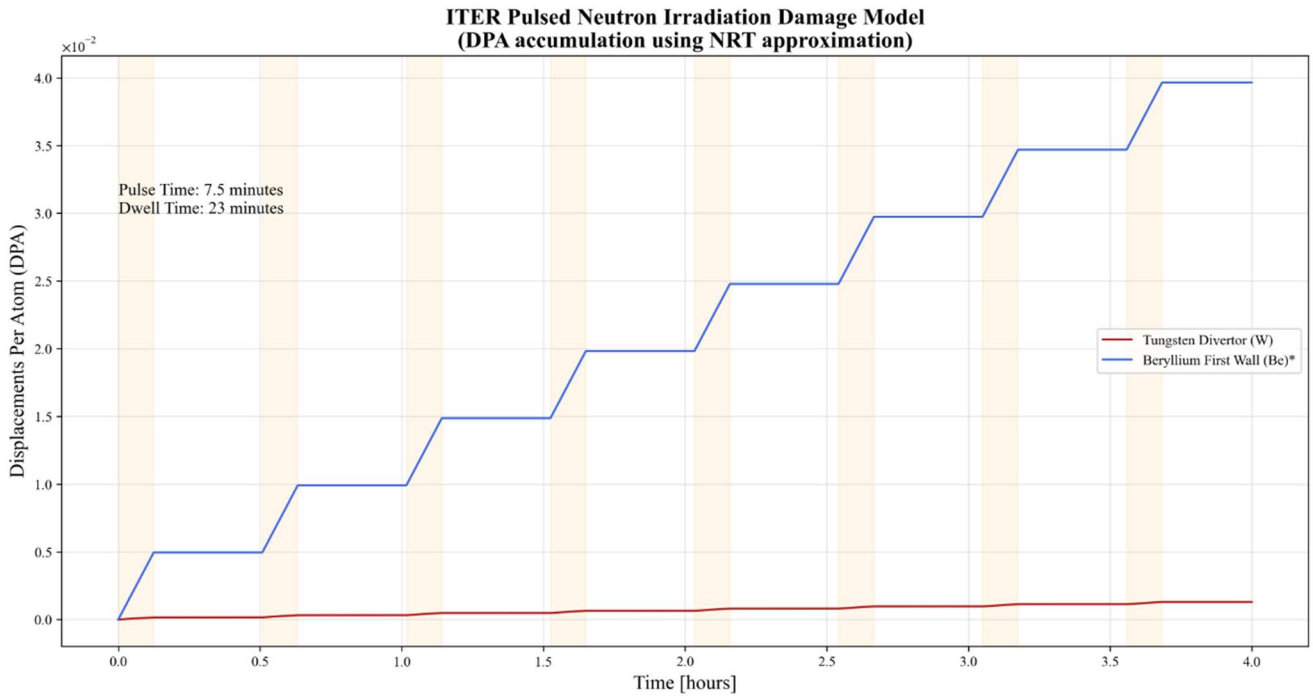
The results show that beryllium accumulates a significantly higher dpa than tungsten over the same irradiation period. This is because the equation for energy transfer

$$\frac{4m_n m_A}{(m_n + m_A)^2}$$

is proportional to $\frac{1}{m_A}$. Since tungsten has a much larger atomic mass than beryllium, the fraction of neutron kinetic energy transferred into lattice recoil is significantly reduced which leads to fewer primary knock-on atoms and lower defect production per interaction.

Furthermore, tungsten has a substantially higher displacement threshold energy than beryllium. This means that a larger amount of transferred energy is required to permanently remove an atom from its lattice site in tungsten. Consequently, even when energy is transferred, fewer atomic displacements are produced compared to beryllium under identical irradiation conditions.

Together, these effects—reduced energy transfer efficiency and higher displacement energy threshold—result in significantly lower dpa accumulation in tungsten relative to beryllium within the same irradiation environment.


ITER Pulsed Neutron Irradiation Damage Model
(DPA accumulation using NRT approximation)
×10⁻²
Pulse Time: 7.5 minutes
Dwell Time: 23 minutes
Tungsten Divertor (W)
Beryllium First Wall (Be)*
Displacements Per Atom (DPA)
Time [hours]
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0